DMM vs. PCA

A comparison

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Contents

1 Introduction

In this document we want to compare the effect of the PCA versus our proposal regarding the construction of the DMM. For this purpose, we will apply both methods to a random dataset, considering:

- PCA
- DMM
- DMM without the square root

2 Dataset

We must have columns with constant sum. We will generate a random dataset with 5 columns of zeros and ones, using as a reference the iris dataset, with only 2 species and 2 variables, which will be discretized.

```
## NULL
```

```
## # A tibble: 3 x 6
##     SL1   SL2   SW2   PL2   PW2 Class
##   <dbl> <dbl> <dbl> <dbl> <dbl> <fct>
## 1     0     1     1     0     0 1
## 2     1     0     1     0     0 1
## 3     0     1     1     0     0 1
```

```
## # A tibble: 3 x 6
##     SL1   SL2   SW2   PL2   PW2 Class
##   <dbl> <dbl> <dbl> <dbl> <dbl> <fct>
## 1     0     1     1     0     0 1
## 2     1     0     1     0     0 1
## 3     0     1     1     0     0 1
```

```
## # A tibble: 3 x 6
##      SL1    SL2   SW2    PL2    PW2 Class
##    <dbl>  <dbl> <dbl>  <dbl>  <dbl> <fct>
## 1 -1.36   1.36    1.1 -0.918 -0.918 1
## 2  0.730 -0.730   1.1 -0.918 -0.918 1
## 3 -1.36   1.36    1.1 -0.918 -0.918 1
```

2.1 Data description

Notice that all variables are dummified, so we are working with the following variables: `SL1`, `SL2`, `SW2`, `PL2`, `PW2`, `Class`, obtained from the original dataset `iris`. Notice that we only perform 2 cuts to improve the visualization, even if that reduces our accuracy.

```
## tibble [100 x 6] (S3: tbl_df/tbl/data.frame)
##  $ SL1  : num [1:100] 0 1 0 1 0 1 1 1 0 1 ...
##  $ SL2  : num [1:100] 1 0 1 0 1 0 0 0 1 0 ...
##  $ SW2  : num [1:100] 1 1 1 0 0 0 1 0 0 0 ...
##  $ PL2  : num [1:100] 0 0 0 0 0 0 0 0 0 0 ...
##  $ PW2  : num [1:100] 0 0 0 0 0 0 0 0 0 0 ...
##  $ Class: Factor w/ 2 levels "1","2": 1 1 1 1 1 1 1 1 1 1 ...
```

2.2 Visualization

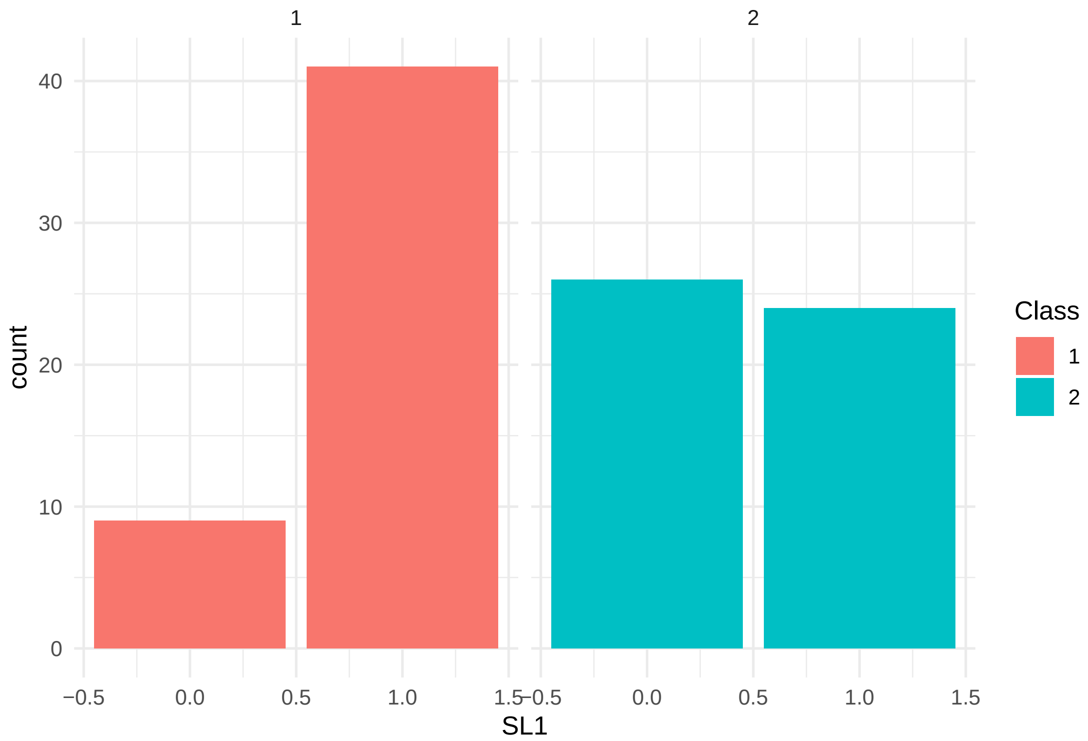

Figure 1: 2 barplots for SL1

3 DMM

Beware, for at some point during the computation of DMM, in particular at `DMM %*% U_2`, we obtain the same values time and time again:

```
## .
##  0.505436206853426  0.685707578831756 -0.804983889505483
##                 31                 31                 20
```

> **Warning!** We are obtaining the same representation for 13 different points... but why?

The previous example explains the following plot:

We can add some jitter to it for a better visualization:

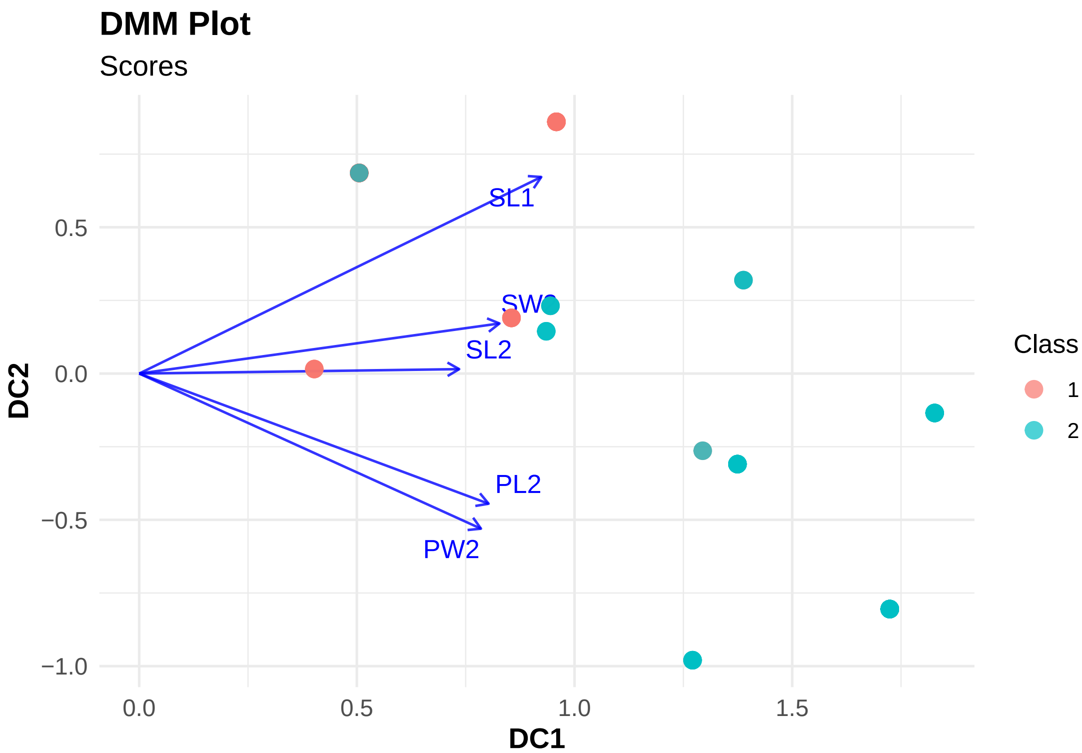

Figure 2: DMM plot: 2D representation

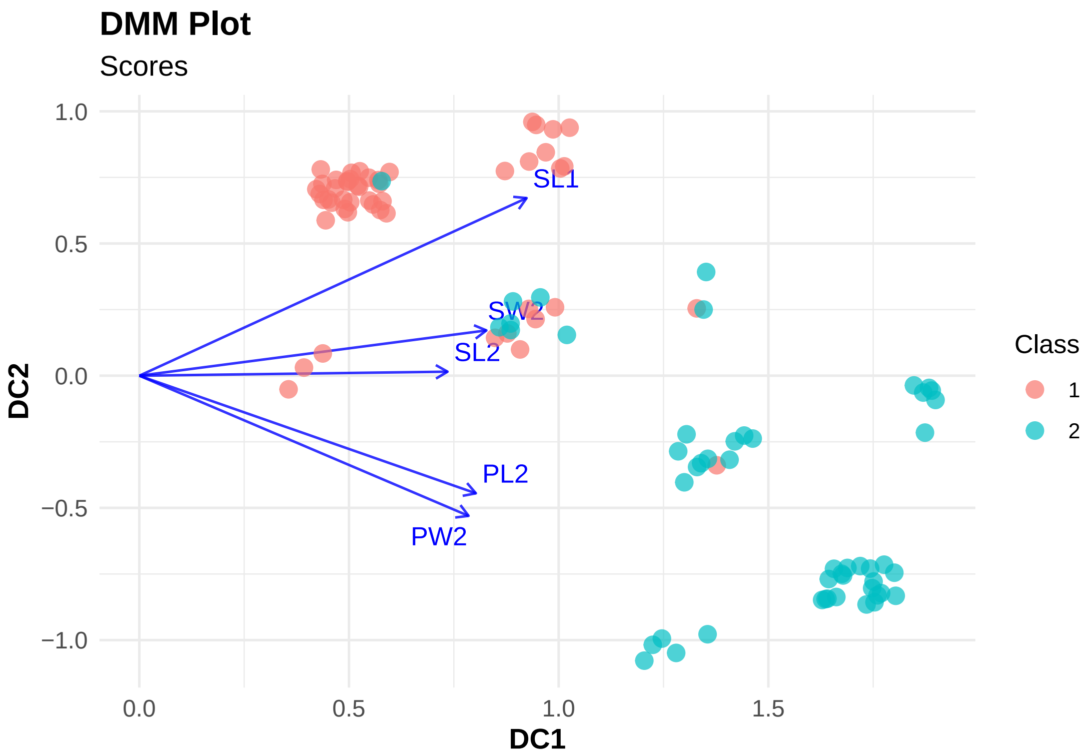

Figure 3: DMM plot: 2D representation with jitter

4 PCA

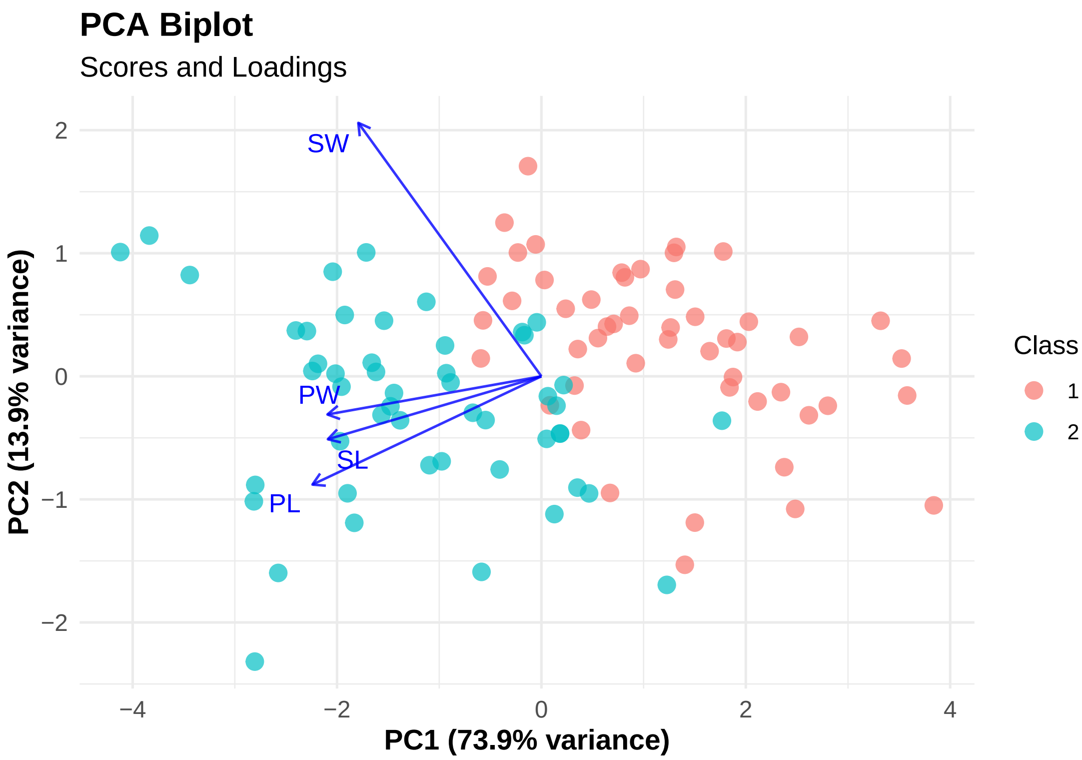

Figure 4: PCA plot: 2D representation

5 TODO: DMM alternative